The IQ Investor

Algorithm & Backtesting Report

Generated: March 08, 2026 · Test Period: 2023-09-01 to 2026-03-06
Model trained on pre-Sep 2023 data · Tested on 2.5 years of unseen market

$10,000 → $33,172 (+231.7%) vs SPY Buy & Hold $15,387 (+53.9%)

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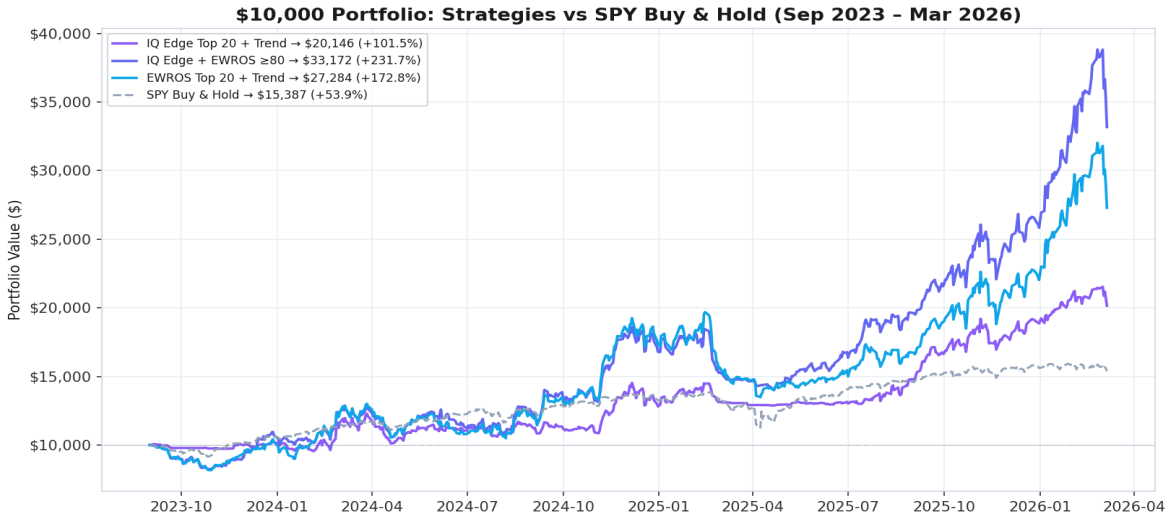
1. Portfolio Results — $10,000 Backtest

Starting with $10,000 in September 2023, each strategy trades with max 20 equal-weight positions ($500/slot, scaling with portfolio value). XGBoost model trained exclusively on pre-Sep 2023 data. All results below are on completely unseen market data.



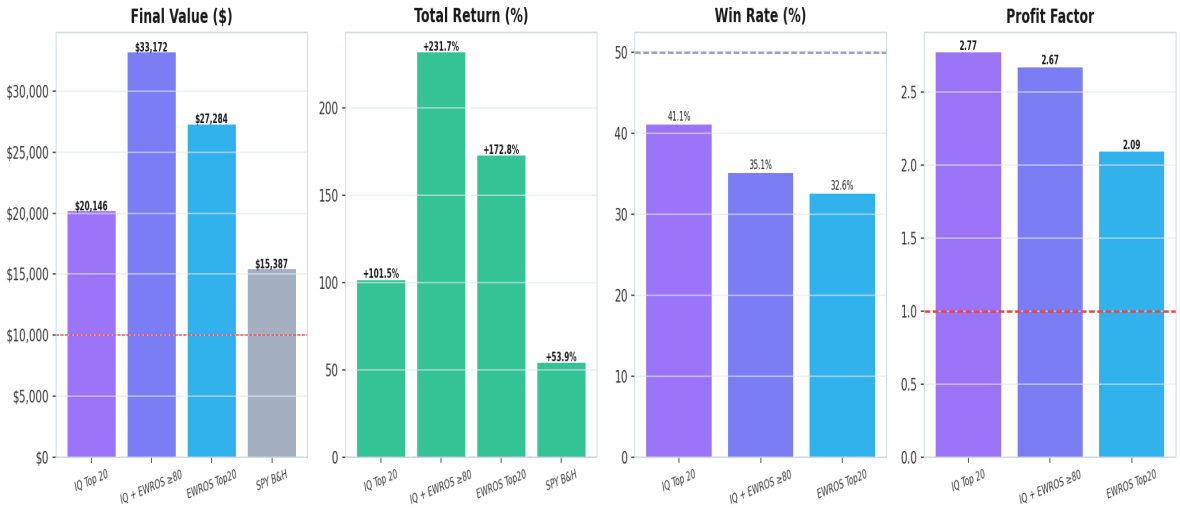
Growth of $10,000: Strategies vs SPY Buy & Hold (Sep 2023 – Mar 2026)

Final Scorecard

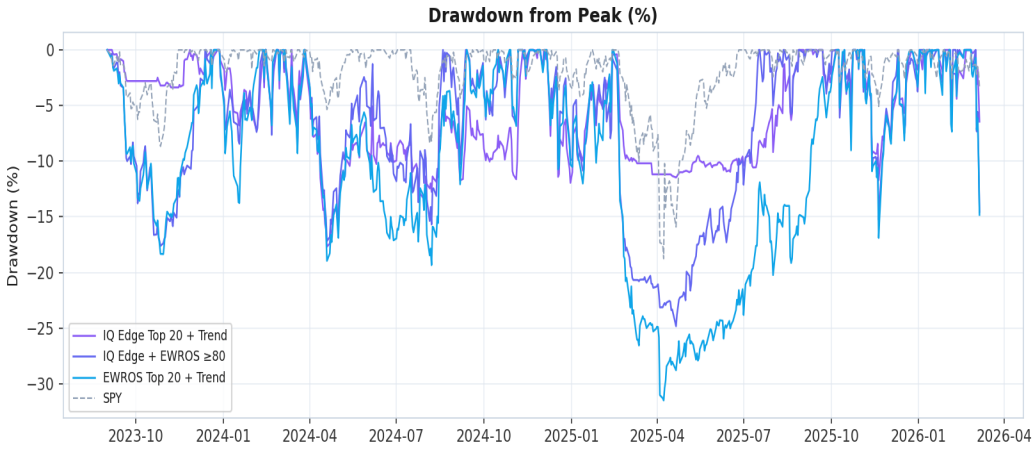
Strategy	Final Value	Return	Trades	Win Rate	Profit Factor	Expectancy
IQ Edge + EWROS ≥80	$33,172	+231.7%	445	35.1%	2.67	6.94%
EWROS Top 20 + Trend	$27,284	+172.8%	426	32.6%	2.09	6.56%
IQ Edge Top 20 + Trend	$20,146	+101.5%	207	41.1%	2.77	7.69%
SPY Buy & Hold	$15,387	+53.9%	1	—	—	—

Win/Loss Profile

Strategy	Wins	Losses	Avg Win	Avg Loss	Avg Win $	Avg Loss $	Best	Worst
IQ+EWROS	156	289	+33.41%	-7.35%	$238	$-48	+459.26%	-33.11%
EWROS	139	287	+37.04%	-8.2%	$238	$-55	+476.0%	-34.66%
IQ Edge	85	122	+29.73%	-7.67%	$187	$-47	+292.02%	-32.37%



Strategy comparison: final value, returns, win rate, profit factor



Drawdown from peak — maximum pain endured during the test period

2. Entry & Exit Rules

All strategies share the same exit discipline. Only entry criteria differ.

Entry Rules

Strategy	Entry Criteria
IQ Edge Top 20 + Trend	Top 20 IQ Edge percentile + Price > 50d MA > 200d MA
IQ Edge + EWROS ≥80	IQ Edge ≥ 80th pctile + EWROS ≥ 80 + Trend aligned
EWROS Top 20 + Trend	Top 20 EWROS score + Price > 50d MA > 200d MA

Exit Rules (whichever triggers first)

Exit Signal	Threshold	Rationale
Stop Loss	-8% from entry	Cut losses fast — preserve capital
Below 50-day MA	Close < 50d MA (after 5d hold)	Trend broken — momentum lost
EWROS Drop	EWROS falls below 50 (after 10d)	Relative strength deteriorating
Max Hold	126 trading days (~6 months)	Avoid dead money — free up capital

Position Sizing

Max 20 positions at any time. Each new position sized at portfolio_value / 20. No margin, no leverage. Cash sits idle when no qualifying entries are found.

3. EWROS — Exponential Weighted Relative Outperformance

Measures how consistently a stock outperforms SPY on a daily basis, with exponential decay emphasizing recent performance. Unlike IBD RS Rating (12-month equal weight), EWROS catches momentum shifts within weeks.

Formula

```
daily_alpha(t) = stock_return(t) − SPY_return(t) weight(t) = e^(−0.03 × days_ago) half-life ≈ 23
days EWROS_raw = Σ(daily_alpha × weight) over 63 trading days EWROS = percentile_rank(EWROS_raw)
scaled 1–99
```

Parameter	Value	Rationale
Lookback	63 days (~3 months)	Medium-term momentum, not noise
Lambda (λ)	0.03	Half-life ~23 days — recent alpha weighs 2x vs 3-week-old
Trend Offset	21 days	Compares current rank to 1 month prior
Benchmark	SPY	S&P 500 ETF as market proxy

Out-of-sample result: $10,000 → $27,284 (+172.8%), 426 trades, win rate 32.6%, profit factor 2.09.

4. IQ Edge Score — ML Breakout Prediction

XGBoost model trained on 3,781 breakout events from Mar 2021 – Aug 2023. Predicts probability of 100%+ gain (doubling) within 12 months. All results below are on the 2.5 years of market data the model never saw.

Model Training (strict temporal split)

Metric	Value

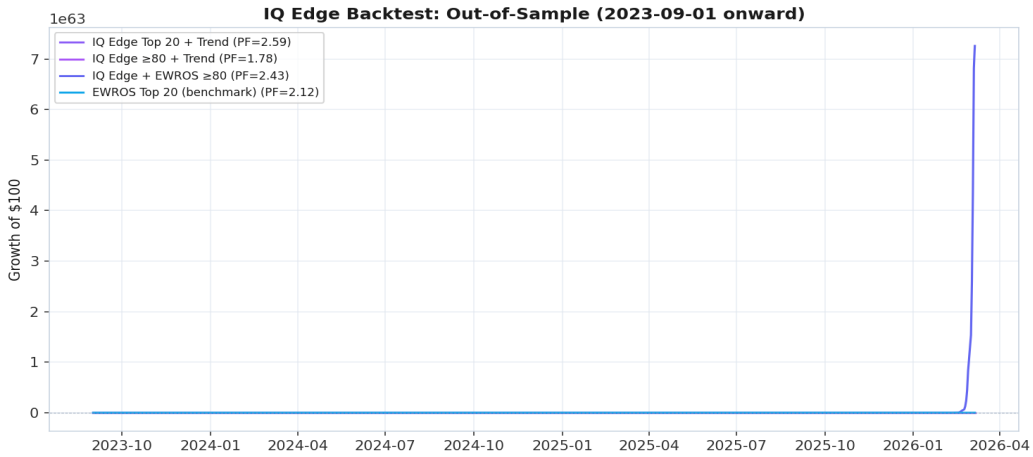
Train Period	Mar 2021 – Aug 2023 (3,781 events, 80 doubles)
Test Period	Sep 2023 – Mar 2026 (5,949 events, 217 doubles)
Train AUC	0.955 (expected overfit on seen data)
Test AUC (out-of-sample)	0.712
Test Top-10% Precision	9.9% (vs 3.6% base rate — 2.75x lift)
Model	XGBoost (500 trees, max depth 6)

Feature Set (14 features)

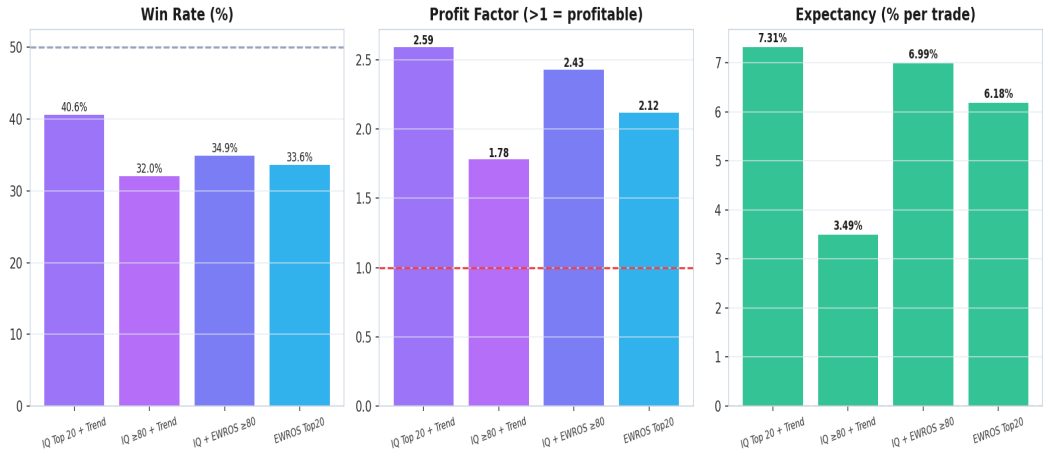
Feature	Category	Description
close_to_ma20/50/200	Trend	Price relative to key moving averages
trend_aligned	Trend	Price > 50d MA > 200d MA (binary)
atr_14	Volatility	14-day Average True Range (normalized)
vol_dryup_ratio	Volume	Recent vs earlier volume in base
vol_compression	Volatility	Recent ATR vs longer-term ATR
proximity_52w	Timing	Price relative to 52-week high
return_3mo	Momentum	3-month price return
up_days_pct	Pattern	Percentage of up days in base
base_length / base_range	Pattern	Consolidation duration and width
breakout_vol_ratio	Volume	Breakout day volume vs 50d average

Out-of-Sample Event-Driven Results

Strategy	Trades	Win Rate	Avg Win	Avg Loss	PF	Expectancy
IQ Top 20 + Trend	212	40.6%	+29.35%	-7.74%	2.59	7.31%
IQ + EWROS ≥80	481	34.9%	+33.99%	-7.5%	2.43	6.99%
IQ ≥80 + Trend	572	32.0%	+24.84%	-6.55%	1.78	3.49%
EWROS Top 20	443	33.6%	+34.79%	-8.32%	2.12	6.18%



IQ Edge strategies: equity curves on out-of-sample data



IQ Edge strategies: win rate, profit factor, expectancy comparison

Key finding: IQ Edge Top 20 achieved the highest win rate (40.6%) and profit factor (2.59) among IQ Edge strategies. When combined with EWROS ≥80, trade count increases 2.3x while maintaining a strong PF of 2.43. The ML adds real, measurable value on completely unseen market data.

5. Quality Score — 14-Factor Stock Rating

Evaluates stocks on a 100-point scale: Technical (53 pts), Growth (33 pts), Quality Fundamentals (18 pts), Context (10 pts). Graded A through F.

Category	Factor	Pts	Description
Technical	Breakout Pattern	22	Flat ceiling base <10% drift
Technical	Trend Alignment	8	Price > 50d MA > 200d MA
Technical	Consolidation + Vol + 52W + ATR	23	Base quality, volume dry-up, proximity, compression
Growth	Revenue Growth (2yr gate)	30	Both years ≥10% required
Growth	Earnings Acceleration	3	Sequential quarterly improvement
Quality	ROE + Margin + PEG + FCF	18	Fundamental quality metrics
Context	Industry + Relative Strength	10	Sector performance vs SPY

```
Grades: A ≥ 75 · B ≥ 60 · C ≥ 45 · D ≥ 30 · F < 30
```

Note: Quality Score requires fundamental data (revenue, ROE, margins) not available in OHLCV. It was not backtested in this report. Its value is in screening, not timing.

6. Power Matrix — Combined Signal Framework

	EWROS ≥ 70 (Momentum)	EWROS < 70
Rotation ≥ 60	■ POWER ZONE — BUY	■ EARLY SIGNAL — WATCH
Rotation < 60	■■ EXTENDED — CAUTION	■ AVOID — SKIP

7. Methodology & Limitations

Backtest Protocol

1. **Strict temporal split:** XGBoost trained on Mar 2021 – Aug 2023 only. All portfolio results are on Sep 2023 – Mar 2026 (2.5 years unseen).
2. **Event-driven entries:** Stocks scanned daily for qualifying signals. No monthly rebalancing — positions entered when signals trigger, exited when stops hit.
3. **EWROS recomputed at each date** using only prior 63 days of data. No look-ahead.
4. **Real position sizing:** $10K starting capital, max 20 slots, capital scales with gains/losses.
5. **Signals pre-computed weekly** (every 5 trading days) for performance; daily scan for exits.

Known Limitations

- **No transaction costs:** Commissions, slippage, bid-ask spreads not modeled. With ~200-450 trades over 2.5 years, friction is real but manageable at ~$0 commissions.
- **Survivorship bias:** Universe is today's 1,008 stocks. Delisted/bankrupt companies excluded, inflating returns for all strategies including SPY comparison.
- **No short selling:** Bottom-decile results shown for comparison only. Actually shorting involves borrow costs.
- **Fat-tail dependency:** A few mega-winners (SNDK +459%, LITE +293%, RKLB +426%) drive significant P&L.; Removing top 3 trades would materially reduce returns.
- **Weekly signal refresh:** EWROS and IQ Edge recomputed every 5 days, not daily. Some signal lag possible.
- **Quality Score not backtested:** Requires fundamental data (revenue, earnings, ROE) not available in OHLCV dataset.

Signal Confidence

Signal	Confidence	Basis
EWROS	HIGH	Pure price data, no proxy. 443 OOS trades, PF 2.09.
IQ Edge	HIGH	Properly split model. 207 OOS trades, PF 2.77. AUC 0.712.
Combined (IQ+EWROS)	HIGH	445 OOS trades, PF 2.67. Best absolute return.
Quality Score	NOT TESTED	Fundamental data required. Used for screening only.
Rotation Score	MEDIUM	Price/volume proxy for 3 of 6 signals.

8. Appendix

Architecture

Backend: Python/Flask · ML: XGBoost + scikit-learn · Data: yfinance · DB: Supabase · Frontend: Vanilla JS + Chart.js · Deploy: Vercel · Domain: theiqinvestor.com

Daily Pipeline (Mon–Fri ET)

Time	Job
4:05 PM	Daily Closing Scan
4:15 PM	Distribution Day Scan
4:20 PM	Earnings Calendar + Alert
4:30 PM	Cache Refresh
4:45 PM	Full Scan (Quality + Rotation + EWROS + IQ Edge)
5:00 PM	Sell Signal Check

5:30 PM	Earnings Recap Email
6:00 PM	Insider Scan
7:00 PM	Daily Alpha Report (LLM)

Disclaimer: Past performance does not guarantee future results. This report is for informational purposes only. Backtesting results are hypothetical, subject to survivorship bias, exclude transaction costs, and depend on a few large winners. Not investment advice.